A CONTINUOUS MAP FROM A QUASI-COMPACT TO A HAUSDORFF SPACE IS UNIVERSALLY CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-topology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The statement that a closed subset of a quasi-compact space is quasi-compact. Invoked inside the counted proof (source lines 3485-3486 and 3490-3491). Makes the fibres and the images of closed subsets quasi-compact.
- The characterization of properness through closedness and quasi-compact fibres. Invoked inside the counted proof (source lines 3487-3488). Reduces universal closedness to closedness plus quasi-compact fibres.
- The statement that the image of a quasi-compact set is quasi-compact and that a quasi-compact subset of a Hausdorff space is closed. Invoked at the end of the counted proof (source lines 3491-3494). Concludes closedness of the image.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-image-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1964–1964. Navigation pointer only: the counted text refers to the Stacks item lemma-image-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:topology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-quasi-compact-in-Hausdorff: sources/stacks-topology/pinned-original-source.tex, source lines 1894–1894. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-in-Hausdorff, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original theorem theorem-characterize-proper: sources/stacks-topology/pinned-original-source.tex, source lines 3370–3370. Navigation pointer only: the counted text refers to the Stacks item theorem-characterize-proper, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces. If X is quasi-compact and Y is Hausdorff, then f is universally closed.*

Proof. Since every point of Y is closed, we see from Lemma lemma-closed-in-quasi-compact that the closed subset $f^{-1}(y)$ of X is quasi-compact for all $y \in Y$. Thus, by Theorem theorem-characterize-proper it suffices to show that f is closed. If $E \subset X$ is closed, then it is quasi-compact (Lemma lemma-closed-in-quasi-compact), hence $f(E) \subset Y$ is quasi-compact (Lemma lemma-image-quasi-compact), hence $f(E)$ is closed in Y (Lemma lemma-quasi-compact-in-Hausdorff). □